

Rayirth Kamath

+91 7972908002 | djxray30@gmail.com

EDUCATION

Sardar Vallabhbhai National Institute of Technology

Surat, Gujarat, India

Bachelor of Technology in Computer Science and Engineering; CGPA: 9.06

2024 – Present

- Relevant Coursework: Data Structures, Design and Analysis of Algorithms, Database Management Systems, Artificial Intelligence

Mushtifund Aryaans Higher Secondary School

Goa, India

Higher Secondary Education (11th, 12th)

2022 – 2024

- JEE Main Rank: 7705; JEE Advanced Rank: 10245; ; IISER Rank: 246

EXPERIENCE

Megalon

Panaji, Goa, India

Software Intern

June, 2026

- Developed secure REST APIs with ASP.NET Core, EF Core (LINQ), FluentValidation, and JWT authentication, for seamless integration with a React-based admin portal.
- Implemented responsive administrative features using React, Redux Toolkit, TanStack Query, and Ant Design.
- Diagnosed and resolved EF Core migration conflicts, synchronized SQL schemas in SSMS.

PROJECTS

Fault-Injection Middleware | *Spring Boot (Webflux), Angular, Java, TypeScript*

- Engineered a reactive, full-stack platform to validate distributed system resilience, using non-blocking interception to synthesize network latency and exceptions
- Developed a real-time telemetry dashboard to visualize Circuit Breaker pattern state transitions for software behavior under controlled degradation

2D Side-Scrolling Engine | *C++, SFML*

- Architected a high-performance 2D simulation from scratch, implementing a custom Entity Component System (ECS) to decouple logic from data and maximize CPU cache coherence
- Applied Data-Oriented Design (DoD) to manage contiguous memory buffers, ensuring deterministic runtime performance for high-density entity rendering

Traffic Simulator | *C++, SFML*

- Designed an agent-based simulation tool to model vehicular flow across directed graph topologies, assessing congestion patterns and throughput efficiency
- Optimized memory management and referential integrity through a strict pointer-based ownership model, eliminating object slicing and ensuring accurate agent-state synchronization

Video Anonymization Pipeline | *Python, PyTorch, OpenCV, YOLOv8*

- Architected an edge-deployed, multi-threaded video pipeline using bounded thread-safe queues and strict I/O timeouts to prevent deadlock and guarantee sub-100ms end-to-end processing latency
- Engineered an offline object detection backend with constant-time pixelation redaction, dynamically profiling hardware capabilities (CUDA/CPU) to scale inference resolution and maintain deterministic 30 FPS throughput

TECHNICAL SKILLS

Languages: C++, Java, C#, C, TypeScript, JavaScript, Python, SQL, HTML/CSS, Kotlin, Go

Tools: Git, GitHub, VS Code, IntelliJ IDEA, Visual Studio, Vim